Pedestrian Intention and Trajectory Prediction in Unstructured Traffic Using IDD-PeD

Ruthvik Bokkasam[1], Shankar Gangisetty[1], A. H. Abdul Hafez[2], and C. V. Jawahar[1]

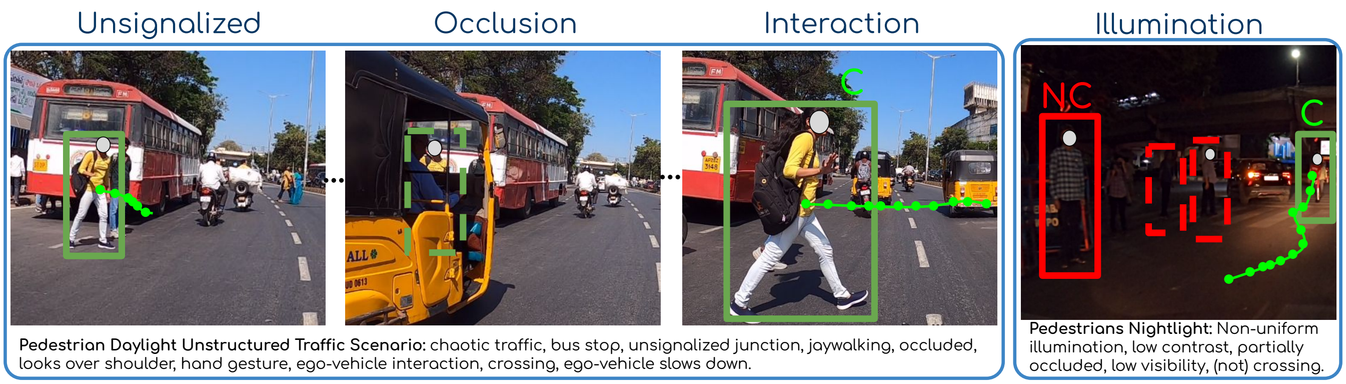

Fig. 1: Illustration of pedestrian intention and trajectory encountering various challenges within our unstructured traffic IDD-PeD dataset. The challenges include occlusions, signalized types, vehicle-pedestrian interactions, and illumination changes. A comparative analysis of these challenges with other datasets is presented in Table I. Intent of C:Crossing with trajectory and NC:Not Crossing.

***Abstract*— With the rapid advancements in autonomous driving, accurately predicting pedestrian behavior has become essential for ensuring safety in complex and unpredictable traffic conditions. The growing interest in this challenge highlights the need for comprehensive datasets that capture unstructured environments, enabling the development of more robust prediction models to enhance pedestrian safety and vehicle navigation. In this paper, we introduce an Indian driving pedestrian dataset designed to address the complexities of modeling pedestrian behavior in unstructured environments, such as illumination changes, occlusion of pedestrians, unsignalized scene types and vehicle-pedestrian interactions. The dataset provides high-level and detailed low-level comprehensive annotations focused on pedestrians requiring the ego-vehicle's attention. Evaluation of the state-of-the-art intention prediction methods on our dataset shows a significant performance drop of up to 15%, while trajectory prediction methods underperform with an increase of up to 1208 MSE, defeating standard pedestrian datasets. Additionally, we present exhaustive quantitative and qualitative analysis of intention and trajectory baselines. We believe that our dataset will open new challenges for the pedestrian behavior research community to build robust models. Project Page: https://cvit.iiit.ac.in/research/projects/cvit-projects/iddped**

I. Introduction

Autonomous vehicles and ADAS must smoothly interact with pedestrians, avoiding collisions, ensuring safe crossings, and promoting efficient movement, especially in dense and dynamic environments [1], [2]. In traffic environments, well-established infrastructure like signals and crosswalks simplify pedestrian behavior prediction. However, the lack of such infrastructure makes pedestrian behavior far more unpredictable [3]. Ego-vehicles must be capable of anticipating sudden changes in pedestrian direction, speed, or intent as real-time predictions of pedestrian intentions and trajectories are critical for enhancing road safety and ensuring smooth navigation in these chaotic and unpredictable traffic settings.

Recent advances in deep learning have significantly improved the perception and prediction of dynamic object motion in structured settings, with several datasets developed for predicting pedestrian intentions and trajectories. Datasets like Argoverse [4], nuScenes [5], IDD-3D [6] and Waymo [7] focus on dynamic agent interactions but are primarily designed for driving perception and planning, not pedestrian intention and navigation. Datasets such as UCY [8], ETH [9], and SSD [10] provide pedestrian trajectories from top-down views, lacking driver-pedestrian interaction insights on the ground. Although vehicle-pedestrian interaction datasets like JAAD [11], PIE [12], TITAN [13], PSI [14], Euro-PVI [15] and TBD [16] target intention and trajectory prediction, they are tailored to structured environments with signals and crosswalks, and do not account for diverse pedestrian behaviors in unstructured settings.

On chaotic urban streets without signals, autonomous vehicles face pedestrian behaviors like jaywalking or sudden

[1]IIIT, Hyderabad, India ruthvik.cvresearch@gmail.com, {shankar.gangisetty@ihub-data., jawahar@}iiit.ac.in
[2]King Faisal University, Al Hofuf, Saudi Arabia aabdulhafiz@kfu.edu.sa

Fig. 2: Rolling behavior is common in unstructured environments, where pedestrians hesitate, pause, or change direction unpredictably, even after they start to cross. *(left)* In our dataset, due to rolling behaviour, the pedestrian does not cross and the ego-vehicle moves ahead, while *(right)* in PIE, the pedestrian crosses since structured settings typically prompt the ego-vehicle to yield giving priority to the pedestrian.

halts. Anticipating these actions is crucial for safe navigation and collision avoidance, as conventional traffic rules often don't apply in such settings. Considering the complexity of pedestrian behavior in unstructured traffic environments, we identify four key factors that impact prediction accuracy: (i) *occlusions*, (ii) *lighting conditions*, (iii) *signalized type*, and (iv) *vehicle-pedestrian interactions*. These factors add unpredictability, demanding robust models to handle such diverse scenarios. However, most existing datasets overlook these nuances, limiting their effectiveness in modeling pedestrian behavior in these challenging conditions.

Now, let us understand these four key factors that are crucial within unstructured traffic conditions, see scenarios in Fig. 1 and comparison in Table I): (i) *Occlusion*: Dense and dynamic traffic often occludes pedestrians, and datasets like TITAN [13] attempt to bypass this by removing occluded regions, which degrades pedestrian intent and trajectory predictions. (ii) *Illumination variation*: The variability in lighting conditions can significantly impact pedestrian behaviour predictions. PIE and TITAN lack nightlight data, while a dataset like NightOwls [17] lacks daylight data, degrading the performance across varying lighting conditions. (iii) *Signalized type*: The absence of traffic signals and crosswalks especially in unstructured settings lead to unpredictable pedestrian-vehicle interactions. JAAD and PIE are captured in scenes with well-established infrastructure including signals and crosswalks simplifying pedestrian behaviour predictions. (iv) *Vehicle-Pedestrian Interactions*: Datasets like PIE and JAAD lack interaction information, which reduces their effectiveness in modeling pedestrian behaviors. In contrast, unstructured environments present more frequent and complex interactions [18]. For example, "rolling behavior" [19] describes pedestrians constantly adjusting their speed and path to navigate traffic, leading to more frequent interactions. As shown in Fig. 2, weak traffic rule enforcement in unstructured environments results in minimal vehicle yielding, even at crosswalks. While TITAN [13] and Euro-PVI [15] capture vehicle-pedestrian interactions, they lack explicit annotations, limiting their ability to model these behaviors accurately.

To bridge these gaps, we introduce the IDD-PeD dataset. The dataset contains over $650K$ annotated bounding boxes and provides comprehensive annotations for pedestrian intention, location, ego-vehicle motion, scene context, occlusion, and pedestrian-vehicle interactions. By capturing these factors, IDD-PeD supports the development of robust models capable of navigating complex, real-world unstructured environments. Fig. 1 depicts samples from the proposed dataset comprising of the diverse challenges and Table I compares our dataset over the existing datasets addressing these challenges. In summary, we make the following contributions:

Dataset	#Pedestrians	#Bounding Boxes	#Annotated Frames	#Behavioral Classes	Traffic Density	OBD	Video Seq.	Group	Location	Occlusion	Nightlight	Signalized	Interaction
JAAD [11]	2.8K	391K	75K	11	Low	✗	✗	✓	✗	✓	✓	✓	✗
PIE [12]	1.8K	750K	293K	11	Low	✓	✓	✗	✗	✓	✗	✓	✗
TITAN [13]	8.6K	NA	75K	33	High	✓	✗	✗	✗	✗	✗	✗	✗
STIP [20]	3.3K	350K	110K	–	Medium	✗	✓	✗	✗	✗	✗*	✗	✗
PSI [14]	180	NA	26K	–	Low	✗	✗	✗	✗	✓	✗*	✗	✗
IDD-PeD (Ours)	5K	686K	205K	19	High	✓	✓	✓	✓	✓	✓	✓	✓

* day and night not explicitly mentioned

TABLE I: Comparison of datasets for pedestrian behavior understanding. On-board diagnostics (OBD) provides ego-vehicle speed, acceleration and GPS information. Group annotation represents the number of pedestrians moving together, in our dataset, about $1,800$ move individually while the rest move in groups of 2 or more. Interaction annotation refers to a label between ego-vehicle and pedestrian, where both influence each other's movements and decisions. ✓and ✗ indicate presence or absence of annotated data.

- We introduce the largest dataset, i.e., IDD-PeD, with more than $650K$ pedestrian bounding boxes and 19 behavioral attributes, captured in dense and unstructured traffic environments. The dataset addresses key challenges such as *Occlusions*, *Lighting Conditions*, *Signalized Types*, and *Vehicle-Pedestrian Interactions* (see Fig. 1 and Table I).
- We benchmark eight baseline methods for pedestrian intention prediction (PIP) and four trajectory prediction (PTP) against the IDD-PeD dataset, showing a performance drop of up to 15% in PIP (see Table II) and an increase in MSE of up to 1208 for deterministic models and 224 for stochastic models in PTP (see Table III) compared to previous benchmark datasets.
- We present a comprehensive quantitative and qualitative analysis of the key challenges highlighting their substantial impact on model robustness (see Tables IV and V).

II. Related Work

A. Pedestrian Intention Prediction

Early works in PIP have evolved from single-frame CNN predictions [11][21] into two main categories: spatio-temporal modeling and feature fusion. Spatio-temporal models use CNNs to extract visual features, followed by RNNs for temporal reasoning [20][22][23], with methods like PIE [12] using ConvLSTMs. Other approaches leverage 3D CNNs, such as 3D DenseNet [24][25], C3D [26], and I3D

[27], or incorporate pose estimation with 2D pose features or Graph Convolutional Networks (GCNs) [28][29].

Adopting feature fusion techniques in recent works has improved prediction accuracy. This process combines various features independently before integrating them. SF-GRU [30] hierarchically fuses features at varying levels of complexity, while PCPA [31] uses RNN encoders and temporal attention, followed by modality attention, to weigh the importance of each feature branch. This allows for the dynamic relevance of information features over time. To address the lack of global scene context in earlier methods, MaskPCPA [32] and [33] integrate segmentation maps with attention-based fusion strategies for a more comprehensive understanding of the environment. Despite the growing focus on transformers, integrating multiple modalities[34][35][36][37][38] for PIP remains challenging due to the need for large training datasets and complex architectures. While transformers show promise, RNN-based feature fusion methods often offer comparable performance with better interpretability, crucial for safety-critical applications.

B. Pedestrian Trajectory prediction

Early works in PTP employ vanilla RNN architectures like PIETraj [12], which estimates crossing intentions which is then used to enhance trajectory predictions. More advanced techniques, such as spatiotemporal graph convolutional networks [39], dynamic-based deep models [40], and knowledge-based intention models [41], improved prediction accuracy. MTNTraj [42] further advanced deterministic prediction by using a multimodal transformer network to capture both high-level and fine-grained motion dynamics.

Recognizing the uncertainty in human motion, recent work has focused toward stochastic models that generate multiple plausible future trajectories. [43] introduced a simulation method for diverse pedestrian trajectories, while BiTraP [44] uses a bi-directional decoder for long-term prediction accuracy and fully attention model presented in [45]. SGNet [46] incorporates goals at multiple temporal scales, further enhancing prediction. These models improve autonomous systems by offering a range of potential future scenarios, as highlighted in [47][48][49].

C. Datasets

Early trajectory datasets like UCY [8], ETH [9], and SSD [10] capture pedestrian movement but use top-down views, making them unsuitable for studying vehicle-pedestrian interactions in ego-view. In contrast, ego-view datasets like nuScenes [5], Waymo [7], and Argoverse2 [4] capture agent interactions but lack detailed pedestrian action annotations. To enhance pedestrian action understanding, datasets like JAAD [11] and STIP [20] introduced binary intent annotations for predicting road-crossing behavior, with PSI [14] adding dynamic intent annotations. However, these datasets often lack detailed ego-vehicle motion and explicit pedestrian action data. PIE [12] and TITAN [13] addressed these gaps by incorporating both, enabling the development of more context-aware models in structured settings.

Despite advancements, existing datasets still lack in capturing complex and challenging scenarios, as illustrated in Fig. 1. Many lack lighting diversity—NightOwls [17] focuses only on nighttime, while PIE and TITAN cover only daytime. Additionally, datasets like Euro-PVI [15] and PePScenes [50] lack explicit pedestrian-vehicle interaction annotations, and there is limited data on occlusions and missing traffic signals or crosswalks in unstructured environments. To address these gaps, we introduce a comprehensive dataset focused on pedestrians in unstructured scenarios which enables the development of robust prediction models for navigating complex, real-world unstructured driving environments.

III. The IDD-PeD Dataset

In this section, we introduce our dataset and present details of the data collection, annotation and data statistics.

A. Data Collection

The dataset is curated from over 100 hours of driving videos captured using either of the two ego-mounted cameras facing the front i.e., GoPro Hero 8 Black at 30 FPS and DDPAI X2SPro dashcam at 25 FPS in HD format. Recordings from the GoPro camera are segmented into video clips of varying lengths, while dashcam recordings are consistently clipped into one-minute segments. We used an OBD sensor synchronized with the cameras which logs ego-motion data like acceleration, speed and GPS at 10 Hz. The data collection was conducted in a densely populated urban area in South Asia, covering various weather conditions. The scenarios captured a wide range of pedestrian behaviors in chaotic, unstructured traffic conditions. This includes both day and night lighting, pedestrian-vehicle interactions, and signalized and unsignalized traffic locations. The scenes vary from commercial roads without pavements to residential areas with narrow streets and no crosswalks, as well as dense traffic scenes with partial or full pedestrian occlusions.

B. Data Annotations

Our dataset provides five types of annotations for pedestrians requiring the ego-vehicle's attention. We used the CVAT tool[1] for manually annotating bounding boxes and other annotations, while 2D pose annotations were automatically extracted using MMPose [51]. We provide examples of the five types of annotations in the supplementary video.

Spatial annotations: The IDD-PeD dataset provides bounding boxes for pedestrians and relevant traffic objects like vehicles, traffic signals, bus stations, and crosswalks, which influence pedestrian movement [19], [52]. It includes $205,145$ annotated frames, with $494,854$ pedestrian boxes and $190,395$ boxes for other traffic objects. All bounding boxes contain occlusion information (see Fig. 3(e)).

Behavioral annotations: are frame-level annotations that capture pedestrian behavior in unstructured environments, providing cues for understanding intentions, predicting actions, and forecasting movements [23], [53]. In our dataset, we categorize behavioral attributes into six distinct classes

[1] https://app.cvat.ai/

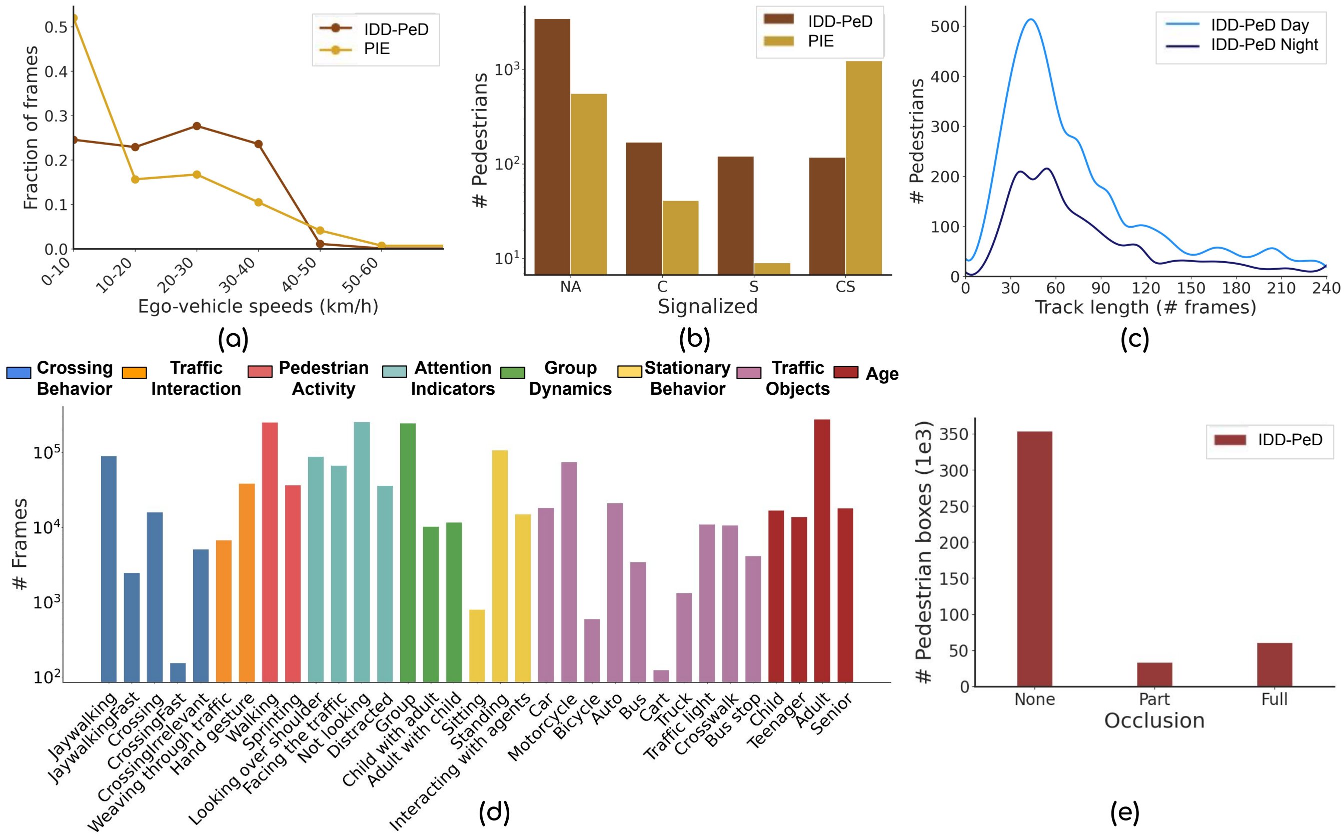

Fig. 3: Annotation instances and data statistics of IDD-PeD. Distribution of (a) frame-level ego-vehicle speeds, (b) pedestrian at signalized types such as crosswalk (C), signal (S), crosswalk and signal (CS), and absence of crosswalk and signal (NA), (c) pedestrians with track lengths at day and night, (d) frame-level different behavior analysis and traffic objects annotation, and (e) pedestrian occlusions.

as shown in Fig. 3(d) comprising of (i) *Crossing Behavior* referring to pedestrian actions when crossing the road. "Crossing" follows traffic rules, like using a crosswalk with a green signal, while "Jaywalking" occurs when crossing illegally [18], such as against a vehicle signal, indicating varying risk levels. (ii) *Traffic Interaction* show pedestrian interactions with vehicles [54], including non-verbal cues like "hand gestures" to signal intentions or "weaving through traffic", where pedestrians navigate between slow-moving vehicles. (iii) *Pedestrian Activity* records pedestrian behaviors when they are moving along the side of the road like "walking", or "sprinting". (iv) *Attention Indicators* capture a pedestrian's focus relative to the ego-vehicle. For instance, "looking over shoulder" signals awareness of the vehicle, while being "distracted" (e.g., using a phone) shows inattention. (v) *Group Dynamics* shows pedestrian behaviors in groups, common in dense areas. For example, a "child with an adult" shows different movement patterns than someone walking alone, while a "crowd" may move as a unit, influencing traffic interactions. (vi) *Stationary Behavior* describes pedestrians staying in one place, often interacting with their surroundings. In busy unstructured streets, this includes engaging with hawkers or others, exemplified by "Communicating with agents" interactions, common in dense urban environments.

Scene annotations: describe the environmental contextual information around the pedestrian, such as the road type (main road, streets), intersection type (mid-block, turns), signalized type (crosswalk, traffic signal), and time of day (day, night). Scene annotations are crucial in unstructured environments as they provide context for understanding chaotic pedestrian movements. By associating behaviors with environmental factors, these annotations help uncover patterns behind random pedestrian actions. For example, in a busy street with pedestrians weaving through traffic, scene annotations can link a pedestrian's sudden change in direction to a nearby bus-stop or a crosswalk, revealing motives behind their movement.

Interaction annotations: capture how pedestrians and vehicles interact in rule-flexible unstructured environments, where both may alter their speed or direction. Each pedestrian track is labeled with a binary indicator showing whether an interaction with the ego-vehicle occurred and a label indicating the nature of the interaction. For example, in a congested intersection, if a pedestrian abruptly changes their path to avoid an approaching vehicle, this interaction is annotated to highlight the challenges in predicting and responding to such dynamic scenarios.

Location annotations: provide spatial context for pedestrian movements in unstructured environments. Attributes such as "near divider", "side of the road", and "near crosswalk" help situate pedestrians within their surroundings. When paired

TABLE II: Evaluation of **PIP baselines** on JAAD, PIE and our datasets.

Model	JAAD [11]		PIE [12]		IDD-PeD (ours)	
	AUC	F_1	AUC	F_1	AUC	F_1
Static [21]	0.75	0.55	0.60	0.41	0.56	0.21
ConvLSTM [55]	0.57	0.32	0.55	0.39	0.50	0.15
C3D [26]	0.81	0.65	0.67	0.52	0.59	0.25
I3D [27]	0.74	0.63	0.73	0.62	0.61	0.30
SF-GRU [30]	0.77	0.53	0.79	0.69	0.59	0.26
SingleRNN [23]	0.77	0.54	0.77	0.67	0.61	0.31
PCPA [31]	0.79	0.56	0.86	0.77	**0.71**	**0.33**
MaskPCPA [32]	0.82	0.63	0.86	0.80	0.61	0.25

with behavior attributes like "looking over shoulder" or "distracted", these location indicators enhance predictions of pedestrian intent. For example, a pedestrian "near crosswalk" who is "distracted" might be at higher risk of crossing unpredictably, improving safety analysis and prediction models.

C. Data Statistics

Fig. 3 gives a detailed overview of the data statistics. Our dataset contains over $5,000$ annotated pedestrians distributed across continuous videos with a varying length of 45 seconds to 10 minutes. Our dataset features an even distribution of ego-vehicle speeds, with a median of 30 km/h (see Fig. 3(a)). In contrast, the PIE dataset includes many frames where the ego-vehicle is stationary, simplifying pedestrian behavior modeling due to the static camera. Pedestrian encounters in our dataset are more frequent in areas without signals or crosswalks (NA), while in structured environments like PIE, presence of crosswalks and traffic signals (CS) leads to smoother pedestrian movement (see Fig. 3(b)). The pedestrian tracks have a median length of approximately 60 frames and cover both day and night scenarios (see Fig. 3(c)). We use the PIE [12] occlusion labeling convention (see Fig. 3(e)). The dataset is split into training (70%) and testing (30%) sets.

IV. Experiments

In this section, we discuss the experimental settings, evaluation of the proposed IDD-PeD dataset on pedestrian intention and trajectory prediction, and further analysis.

A. Experimental Settings

Baseline Methods. We evaluate eight PIP baseline methods on our dataset, namely, Static [21], SingleRNN [23], I3D [27], C3D [26], ConvLSTM [55], SF-GRU [30], PCPA [31], and MaskPCPA [32]. In PTP, we evaluate two deterministic baselines– PIETraj [12], MTN [42] and two stochastic baselines– BiTraP [44], SGNet [46]. These baselines are then evaluated across the key challenges including *occlusions*, *illumination changes*, *unsignalized* scenes, and vehicle-pedestrian *interactions*.

Implementation Details. We implement baseline methods on Intel Xeon E5-2640 v4 processor with multiple Nvidia RTX 2080 Ti GPUs. We fix an observation period of 0.5s and a time-to-event varying between $1-2$s for PIP while in PTP we predict future paths up to 1.5s. For a fair comparison, we adopted the same observation and time settings as in [31] for PIP and [12] for PTP. We use RAFT [56] to extract optical flow information, used by the MTN [42] method. We ensured that for all the crossing cases, both the observation period and the time-to-event precede the pedestrian's road crossing initiation. For methods that use ego-vehicle speed, we report results on PIE and our dataset, since JAAD doesn't provide explicit ego-vehicle speeds.

TABLE III: Evaluation of **PTP baselines** on JAAD, PIE and our datasets. We report MSE results at 1.5s. "-" indicates no results as PIETraj needs explicit ego-vehicle speeds. See supplementary video for MSE with 0.5s and 1s prediction periods.

Method	PIE [12]			JAAD [11]			IDD-PeD (ours)		
	MSE	C-MSE	CF-MSE	MSE	C-MSE	CF-MSE	MSE	C-MSE	CF-MSE
PIETraj [12]	559	520	2162	-	-	-	2181	1979	8960
MTN [42]	**444**	**414**	**1627**	**1005**	**951**	**4010**	**1652**	**1489**	**6668**
BiTraP [44]	102	81	261	222	177	565	338	200	638
SGNet [46]	**88**	**66**	**206**	**197**	**146**	**443**	**310**	**190**	**531**

Evaluation Metrics. We evaluate PIP using AUC and F_1 scores, and PTP using Mean Square Error (MSE), Center MSE (C-MSE), Center Final Mean Square Error (CF-MSE), all measured in squared pixels. For stochastic methods, we report the best-of-20 results, consistent with [57].

B. Intention Prediction Results

In Table II, we report experiments of the PIP baselines on PIE, JAAD and our datasets. We observe that existing models like SF-GRU, SingleRNN, and PCPA are less robust to unstructured environmental changes specifically in the context of our dataset. PCPA shows a superior performance among the baselines with an AUC performance drop of 15% and 8%, and F_1 of 44% and 23%, between our dataset, and PIE and JAAD respectively. Surprisingly, the recent MaskPCPA method, which incorporates global context via segmentation maps while achieving SOTA on PIE and JAAD datasets significantly underperforms on our dataset. One possible reason is the parameter-heavy architecture of PCPA, which has 31M parameters compared to 3M in MaskPCPA. This larger capacity allows PCPA to better model the noisy and complex patterns in unstructured settings.

C. Trajectory Prediction Results

In Table III, we report the MSE results of PTP baselines at various prediction intervals on PIE, JAAD and our datasets. We demonstrate that existing methods such as PIETraj, MTN that are deterministic, BiTraP and SGNet that are stochastic, perform with increased error rates when evaluated on our dataset which indicates a lack of robustness to diverse pedestrian behaviours in unstructured environments. Despite not achieving the best results on our dataset, the stochastic SGNet model has the best overall results, with a performance drop of 222 and 113 MSE between ours and the PIE and JAAD datasets respectively. Similarly, the deterministic MTN has the best results, with a performance drop of $1,208$ and 647 MSE between ours, and PIE and JAAD respectively. The more substantial increase in error rates for PIETraj and MTN on our dataset might be attributed to their heavy reliance on visual features, that could be more susceptible to

TABLE IV: Evaluation of PIP baselines on **Occlusions, Signalized Types, Illumination Changes**, and **Vehicle-Pedestrian Interactions** in our dataset. We report AUC scores.

Method	Occlusion		Signalized		Illumination		Interaction	
	w/o	w	Yes	No	Day	Night	w/o	w
SingleRNN [23]	0.63	0.61	0.70	0.62	0.63	0.58	0.65	0.58
I3D [27]	0.60	0.57	0.48	0.64	0.65	0.51	0.64	0.60
SF-GRU [30]	0.60	0.59	0.60	0.59	0.62	0.53	0.61	0.59
PCPA [31]	**0.72**	**0.71**	**0.66**	**0.72**	**0.74**	**0.65**	**0.73**	**0.69**

TABLE V: Evaluation of PTP baselines on **Occlusions, Signalized Types, Illumination Changes**, and **Vehicle-Pedestrian Interactions** in our dataset. We reported MSE at 1.5s.

Method	Occlusion		Signalized		Illumination		Interaction	
	w/o	w	Yes	No	Day	Night	w/o	w
PIETraj [12]	2055	2658	1434	2374	2190	2294	1561	4127
MTN [42]	1565	1914	1215	1765	1633	1704	1328	2668
BiTraP [44]	301	434	296	341	339	310	249	610
SGNet [46]	**289**	**411**	**281**	**329**	**330**	**301**	**227**	**581**

noise and variability whereas models like SGNet, that use a bounding-box based approach may offer some resilience to the visual complexities of unstructured environments.

D. Further Analysis of Our Dataset Challenges

Here, we examine the diversity and richness of our dataset, with quantitative and qualitative results for PIP and PTP.

Intention Prediction. In Table IV, we report AUC of the best performing models across key challenges present in our dataset. We observe that the performance of all the baselines drops when a part of the pedestrian sequence is *occluded*. While occlusion is important, its impact diminishes when there are enough non-occluded frames in the sequence. We also observe that the *nightlight* performance drops by up to 14% over *daylight* scenes on all baselines making our dataset more challenging. This is attributed to low and in-homogeneous illumination, which hinders the baselines that heavily rely on visual features. Similarly, the presence of *vehicle-pedestrian interactions* significantly impacted the performance of the baselines causing a performance drop of up to 4% compared to situations with no interactions. This can be attributed to a sudden change in the decision of the pedestrian to cross or not cross which may have caused the interaction.

Trajectory Prediction. In Table V, we demonstrate the MSE values at 1.5s across baseline methods, highlighting the challenges within our dataset. The presence of *occlusion* results in a significant MSE increase of 603 and 349 for PIETraj and MTN, respectively, while BiTraP and SGNet show a rise of 133 and 122 MSE, respectively. The performance across *illumination* scenarios is also evident for PIETraj and MTN, which exhibit a performance drop during the nightlight over the daylight. Interestingly, BiTraP and SGNet show less sensitivity to this challenge due to their reliance on bounding boxes rather than raw images. The largest performance drop comes from scenarios involving *pedestrian-vehicle interactions* with a drop by up to 354 and $1,340$ MSE for SGNet and MTN. Interactions often involve instantaneous movements by pedestrians and modelling this is critical in unstructured scenarios. Finally, the presence of *signalized* infrastructure correlates with improved model performance in trajectory prediction that can be attributed to fewer interactions and smoother pedestrian flow.

Qualitative Analysis. We illustrate qualitative evaluation of the best (SGNet) and the worst (PIETraj) PTP models on our dataset in Fig. 4. *Interaction:* As the ego-vehicle approaches the crossing pedestrian, the pedestrian exhibits movement away from the ego-vehicle, a behavior both SGNet and PIETraj fail to capture, likely due to the presence of a crosswalk. *Nightlight:* PIETraj does not predict the pedestrian's crossing, possibly due to low visibility and uneven lighting. *Partial occlusion:* Both models struggle to predict the pedestrian's future location, likely because of occlusion during the observation period. *Signalized type:* Both models perform well in signalized settings, where smooth motion and the ego-vehicle at rest simplify prediction.

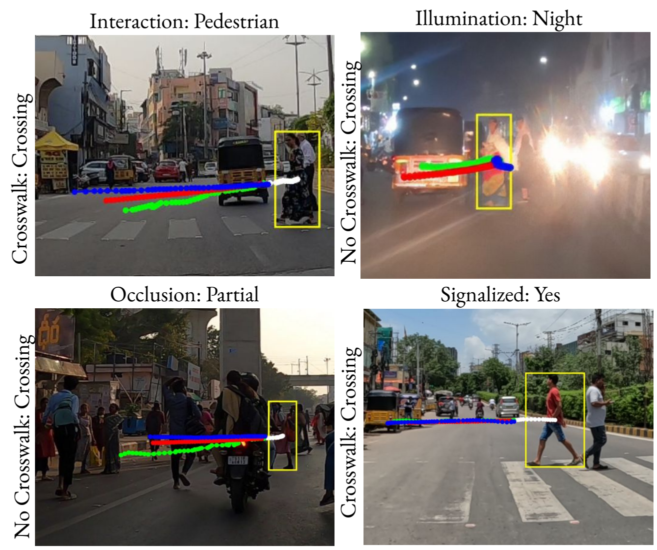

Fig. 4: Qualitative evaluation of the best and worst PTP models on our dataset. Red: SGNet, Blue: PIETraj, Green: Ground truth, White: Observation period. To better illustrate and highlight key factors in PIP and PTP methods, a qualitative analysis will be provided in the supplementary video.

V. Conclusions

In this work, we introduced IDD-PeD, a large pedestrian dataset with detailed behavioral annotations designed for unstructured traffic challenges, including occlusions, signalized types, vehicle-pedestrian interactions, and illumination variations. Extensive experiments on state-of-the-art pedestrian intention and trajectory prediction baseline methods on our dataset higlights its complexity, with both quantitative and qualitative evaluations revealing significant performance degradation compared to existing datasets.

This dataset provides key insights into vehicle-pedestrian interactions, aiding autonomous systems in anticipating pedestrian behavior. Combined with technological advancements, it aims to enhance navigation and ensure safety in unstructured and unpredictable traffic environments.

REFERENCES

[1] K. D. Katyal, G. D. Hager, and C.-M. Huang, "Intent-aware pedestrian prediction for adaptive crowd navigation," in *2020 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 3277–3283, 2020.

[2] S. Liu, P. Chang, Z. Huang, N. Chakraborty, K. Hong, W. Liang, D. L. McPherson, J. Geng, and K. Driggs-Campbell, "Intention aware robot crowd navigation with attention-based interaction graph," in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 12015–12021, 2023.

[3] C. Parikh, R. Saluja, C. V. Jawahar, and R. Kiran Sarvadevabhatla, "Idd-x: A multi-view dataset for ego-relative important object localization and explanation in dense and unstructured traffic," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 14815–14821, 2024.

[4] B. Wilson, W. Qi, T. Agarwal, J. Lambert, J. Singh, S. Khandelwal, B. Pan, R. Kumar, A. Hartnett, J. K. Pontes, D. Ramanan, P. Carr, and J. Hays, "Argoverse 2: Next generation datasets for self-driving perception and forecasting," in *NeurIPS Datasets and Benchmarks*, 2021.

[5] H. Caesar, V. Bankiti, A. H. Lang, S. Vora, V. E. Liong, Q. Xu, A. Krishnan, Y. Pan, G. Baldan, and O. Beijbom, "nuscenes: A multimodal dataset for autonomous driving," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 11621–11631, 2020.

[6] S. Dokania, A. H. A. Hafez, A. Subramanian, M. Chandraker, and C. Jawahar, "Idd-3d: Indian driving dataset for 3d unstructured road scenes," in *2023 IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 4471–4480, 2023.

[7] P. Sun, H. Kretzschmar, X. Dotiwalla, A. Chouard, V. Patnaik, P. Tsui, J. Guo, Y. Zhou, Y. Chai, B. Caine, *et al.*, "Scalability in perception for autonomous driving: Waymo open dataset," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 2446–2454, 2020.

[8] A. Lerner, Y. Chrysanthou, and D. Lischinski, "Crowds by example," in *Computer graphics forum*, vol. 26, pp. 655–664, Wiley Online Library, 2007.

[9] S. Pellegrini, A. Ess, K. Schindler, and L. V. Gool, "You'll never walk alone: Modeling social behavior for multi-target tracking," in *ICCV*, pp. 261–268, IEEE Computer Society, 2009.

[10] A. Robicquet, A. Sadeghian, A. Alahi, and S. Savarese, "Learning social etiquette: Human trajectory understanding in crowded scenes," in *Computer Vision–ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11-14, 2016, Proceedings, Part VIII 14*, pp. 549–565, Springer, 2016.

[11] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Are they going to cross? a benchmark dataset and baseline for pedestrian crosswalk behavior," in *Proceedings of the IEEE International Conference on Computer Vision Workshops*, pp. 206–213, 2017.

[12] A. Rasouli, I. Kotseruba, T. Kunic, and J. K. Tsotsos, "Pie: A large-scale dataset and models for pedestrian intention estimation and trajectory prediction," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 6262–6271, 2019.

[13] S. Malla, B. Dariush, and C. Choi, "TITAN: Future forecast using action priors," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 11186–11196, 2020.

[14] T. Chen, T. Jing, R. Tian, Y. Chen, J. Domeyer, H. Toyoda, R. Sherony, and Z. Ding, "Psi: A pedestrian behavior dataset for socially intelligent autonomous car," *arXiv preprint arXiv:2112.02604*, 2021.

[15] A. Bhattacharyya, D. O. Reino, M. Fritz, and B. Schiele, "Euro-pvi: Pedestrian vehicle interactions in dense urban centers," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 6408–6417, 2021.

[16] A. Wang, D. Sato, Y. Corzo, S. Simkin, A. Biswas, and A. Steinfeld, "Tbd pedestrian data collection: Towards rich, portable, and large-scale natural pedestrian data," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 637–644, 2024.

[17] L. Neumann, M. Karg, S. Zhang, C. Scharfenberger, E. Piegert, S. Mistr, O. Prokofyeva, R. Thiel, A. Vedaldi, A. Zisserman, *et al.*, "Nightowls: A pedestrians at night dataset," in *Computer Vision–ACCV 2018: 14th Asian Conference on Computer Vision, Perth, Australia, December 2–6, 2018, Revised Selected Papers, Part I 14*, pp. 691–705, Springer, 2019.

[18] K. Mukoya, E. Weng, R. Choudhury, and K. Kitani, "Jaywalkervr: A vr system for collecting safety-critical pedestrian-vehicle interactions," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 9600–9607, 2024.

[19] V. Vasudevan, M. Mehta, and B. Dutta, "Pedestrian temporal gap acceptance behavior at unsignalized intersections in kanpur, india," *Transportation research part F: traffic psychology and behaviour*, vol. 74, pp. 95–103, 2020.

[20] B. Liu, E. Adeli, Z. Cao, K.-H. Lee, A. Shenoi, A. Gaidon, and J. C. Niebles, "Spatiotemporal relationship reasoning for pedestrian intent prediction," 2020.

[21] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," *arXiv preprint arXiv:1409.1556*, 2014.

[22] J. Lorenzo, I. Parra, F. Wirth, C. Stiller, D. F. Llorca, and M. A. Sotelo, "Rnn-based pedestrian crossing prediction using activity and pose-related features," in *2020 IEEE Intelligent Vehicles Symposium (IV)*, pp. 1801–1806, IEEE, 2020.

[23] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, "Do they want to cross? understanding pedestrian intention for behavior prediction," in *2020 IEEE Intelligent Vehicles Symposium (IV)*, pp. 1688–1693, IEEE, 2020.

[24] K. Saleh, M. Hossny, and S. Nahavandi, "Real-time intent prediction of pedestrians for autonomous ground vehicles via spatio-temporal densenet," in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 9704–9710, IEEE, 2019.

[25] K. Saleh, M. Hossny, and S. Nahavandi, "Spatio-temporal densenet for real-time intent prediction of pedestrians in urban traffic environments," *Neurocomputing*, vol. 386, pp. 317–324, 2020.

[26] D. Tran, L. Bourdev, R. Fergus, L. Torresani, and M. Paluri, "Learning spatiotemporal features with 3d convolutional networks," in *Proceedings of the IEEE international conference on computer vision*, pp. 4489–4497, 2015.

[27] J. Carreira and A. Zisserman, "Quo vadis, action recognition? a new model and the kinetics dataset," in *proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*, pp. 6299–6308, 2017.

[28] M. Sucha, D. Dostal, and R. Risser, "Pedestrian-driver communication and decision strategies at marked crossings," *Accident Analysis & Prevention*, vol. 102, pp. 41–50, 2017.

[29] P. R. G. Cadena, M. Yang, Y. Qian, and C. Wang, "Pedestrian graph: Pedestrian crossing prediction based on 2d pose estimation and graph convolutional networks," in *2019 IEEE Intelligent Transportation Systems Conference (ITSC)*, pp. 2000–2005, IEEE, 2019.

[30] A. Rasouli, I. Kotseruba, and J. K. Tsotsos, "Pedestrian action anticipation using contextual feature fusion in stacked rnns," *arXiv preprint arXiv:2005.06582*, 2020.

[31] I. Kotseruba, A. Rasouli, and J. K. Tsotsos, "Benchmark for evaluating pedestrian action prediction," in *Proceedings of the IEEE/CVF winter conference on applications of computer vision*, pp. 1258–1268, 2021.

[32] D. Yang, H. Zhang, E. Yurtsever, K. A. Redmill, and Ü. Özgüner, "Predicting pedestrian crossing intention with feature fusion and spatio-temporal attention," *IEEE Transactions on Intelligent Vehicles*, vol. 7, no. 2, pp. 221–230, 2022.

[33] A. Rasouli, T. Yau, M. Rohani, and J. Luo, "Multi-modal hybrid architecture for pedestrian action prediction," in *2022 IEEE intelligent Vehicles symposium (IV)*, pp. 91–97, IEEE, 2022.

[34] S. Zhao, H. Li, Q. Ke, L. Liu, and R. Zhang, "Action-vit: Pedestrian intent prediction in traffic scenes," *IEEE Signal Processing Letters*, vol. 29, pp. 324–328, 2021.

[35] L. Achaji, J. Moreau, T. Fouqueray, F. Aioun, and F. Charpillet, "Is attention to bounding boxes all you need for pedestrian action prediction?," in *2022 IEEE Intelligent Vehicles Symposium (IV)*, pp. 895–902, IEEE, 2022.

[36] Y. Zhou, G. Tan, R. Zhong, Y. Li, and C. Gou, "Pit: Progressive interaction transformer for pedestrian crossing intention prediction," *IEEE Transactions on Intelligent Transportation Systems*, 2023.

[37] J. Lorenzo, I. P. Alonso, R. Izquierdo, A. L. Ballardini, Á. H. Saz, D. F. Llorca, and M. Á. Sotelo, "Capformer: Pedestrian crossing action prediction using transformer," *Sensors*, vol. 21, no. 17, p. 5694, 2021.

[38] Z. Zhang, R. Tian, and Z. Ding, "Trep: Transformer-based evidential prediction for pedestrian intention with uncertainty," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 37, pp. 3534–3542, 2023.

[39] S. Liu, H. Liu, Y. Wang, J. Sun, and T. Mao, "Mdst-dgcn: A multilevel dynamic spatiotemporal directed graph convolutional network

for pedestrian trajectory prediction," *Computational intelligence and neuroscience*, vol. 2022, no. 1, p. 4192367, 2022.

[40] H. Wang, W. Zhi, G. Batista, and R. Chandra, "Pedestrian trajectory prediction using dynamics-based deep learning," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 15068–15075, IEEE, 2024.

[41] J. Zhou, X. Bai, W. Fu, B. Ning, and R. Li, "Pedestrian intention estimation and trajectory prediction based on data and knowledge-driven method," *IET Intelligent Transport Systems*, vol. 18, no. 2, pp. 315–331, 2024.

[42] Z. Yin, R. Liu, Z. Xiong, and Z. Yuan, "Multimodal transformer networks for pedestrian trajectory prediction.," in *IJCAI*, pp. 1259–1265, 2021.

[43] C. Anderson, X. Du, R. Vasudevan, and M. Johnson-Roberson, "Stochastic sampling simulation for pedestrian trajectory prediction," in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 4236–4243, IEEE, 2019.

[44] Y. Yao, E. Atkins, M. Johnson-Roberson, R. Vasudevan, and X. Du, "Bitrap: Bi-directional pedestrian trajectory prediction with multi-modal goal estimation," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 1463–1470, 2021.

[45] A. Rasouli, "A novel benchmarking paradigm and a scale-and motion-aware model for egocentric pedestrian trajectory prediction," in *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 5630–5636, IEEE, 2024.

[46] C. Wang, Y. Wang, M. Xu, and D. J. Crandall, "Stepwise goal-driven networks for trajectory prediction," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 2716–2723, 2022.

[47] D. Meng, G. Zhao, and F. Yan, "Social-stgmlp: A social spatio-temporal graph multi-layer perceptron for pedestrian trajectory prediction," *Information*, vol. 15, no. 6, p. 341, 2024.

[48] N. Uhlemann, F. Fent, and M. Lienkamp, "Evaluating pedestrian trajectory prediction methods with respect to autonomous driving," *IEEE Transactions on Intelligent Transportation Systems*, 2024.

[49] H. Alghodhaifi and S. Lakshmanan, "Holistic spatio-temporal graph attention for trajectory prediction in vehicle–pedestrian interactions," *Sensors*, vol. 23, no. 17, p. 7361, 2023.

[50] A. Rasouli, T. Yau, P. Lakner, S. Malekmohammadi, M. Rohani, and J. Luo, "Pepscenes: A novel dataset and baseline for pedestrian action prediction in 3d," *arXiv preprint arXiv:2012.07773*, 2020.

[51] M. Contributors, "Openmmlab pose estimation toolbox and benchmark." `https://github.com/open-mmlab/mmpose`, 2020.

[52] W. Zhang, H. Guo, C. Wang, K. Wang, W. Huang, Q. Xu, H. Tang, B. Yang, and R. Yan, "Analysis of pedestrian illegal crossing at unmarked segments: environmental factors, pedestrian characteristics and crossing behaviours," *Transportation research part F: traffic psychology and behaviour*, vol. 99, pp. 339–355, 2023.

[53] Y. Yao, E. Atkins, M. J. Roberson, R. Vasudevan, and X. Du, "Coupling intent and action for pedestrian crossing behavior prediction," *arXiv preprint arXiv:2105.04133*, 2021.

[54] G. M. Muktadir and J. Whitehead, "Adaptive pedestrian agent modeling for scenario-based testing of autonomous vehicles through behavior retargeting," in *IEEE Int. Conf. Robot. Automat.(ICRA)*, 2024.

[55] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-c. Woo, "Convolutional lstm network: A machine learning approach for precipitation nowcasting," *Advances in neural information processing systems*, vol. 28, 2015.

[56] Z. Teed and J. Deng, "RAFT: Recurrent all-pairs field transforms for optical flow," in *Computer Vision–ECCV 2020: 16th European Conference, Glasgow, UK, August 23–28, 2020, Proceedings, Part II 16*, pp. 402–419, Springer, 2020.

[57] A. Gupta, J. Johnson, L. Fei-Fei, S. Savarese, and A. Alahi, "Social gan: Socially acceptable trajectories with generative adversarial networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 2255–2264, 2018.